



Deflagration-to-detonation transition in hot HMX and HMX-based polymer-bonded explosives

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ABSTRACT

The deflagration-to-detonation transition (DDT) in hot, thermally damaged HMX (δ -phase) and HMX-based polymer-bonded explosives (PBX 9501, LX-14, LX-10 and PBX 9012) differs in some respects from what has been observed in similar tests (DDT tube experiments) with room temperature granular explosives. We provide streak images with other observations and demonstrate the behavior can be binned according to the degree of porosity evolved from physical and chemical damage to the compositions. In each bin, the DDT behavior eventually organizes to resemble Type I DDT, but differs in the early-stage burn phenomena and how a propulsive thermal explosion event arises. We argue that the hot explosive properties, such as permeability and compressibility, and the morphological characteristics of the thermal damage, control the physical mechanism for establishment of the thermal explosion. In some cases, these observations may need to be carefully implemented in numerical models to increase predictive value for these materials at elevated temperature. With high-porosity ($\phi \approx 20$ –50%), the PBXs behaved like granular beds. At intermediate levels of porosity ($\phi \approx 4$ –20%), the transition occurred over longer distances and the process is characterized by a weak, slow convective burn precursor that sets up thermal runaway to explosion behind the flame infiltration front. In the low-porosity bin ($\phi \approx 0$ –4%), where run lengths were short, the thermal explosion is the result of compressive, and possibly also, deconsolidative burning. It is clear that the high-temperature conditions imposed in these experiments caused sensitization towards DDT and this effect was most apparent in tests where the explosive was heated until runaway and auto-ignition where transition distances were among the shortest measured. Practical findings include there being some safety benefit for having binders in HMX formulations, especially when the binder is thermally stable.

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1. Introduction

There has been a considerable amount of research done to describe and model the physics of deflagration-to-detonation transition (DDT) in granular forms of explosives and propellants, at room temperature. However, few of these efforts were directed towards understanding the evolution of reaction violence in high-initial-density pressings of polymer-bonded explosives (PBXs) following thermal insult and cookoff. The assumption has traditionally been that the mechanistic description of DDT derived from experiments performed with beds of powder at various densities would apply to other common solid explosive forms; e.g. PBXs. This implies that

all porosity, regardless of morphological details, scale and spatial distribution, has the same effect on the mechanics of DDT.

The purpose of the work in this paper is to examine the propensity for DDT in HMX¹-based PBXs before and during thermal exposure. It examines the *in situ* response of several practical explosive compositions ignited in a state of thermal runaway or following auto-ignition. We seek to evaluate the assumption that mechanisms derived from traditional DDT tube experiments with powders capture the behavior of PBXs, and if not, understand reasons for differences. This work is focused on the experimental methods and results from hot DDT tube experiments performed with HMX powders and PBXs: PBX 9501, LX-14, LX-10 and PBX 9012 (see Table 2). We will discuss trends in the data and features that we believe to be important in differentiating the response of

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¹ Octahydro-1,3,5,7-tetranitro-1,3,5,7-tetrazocine.

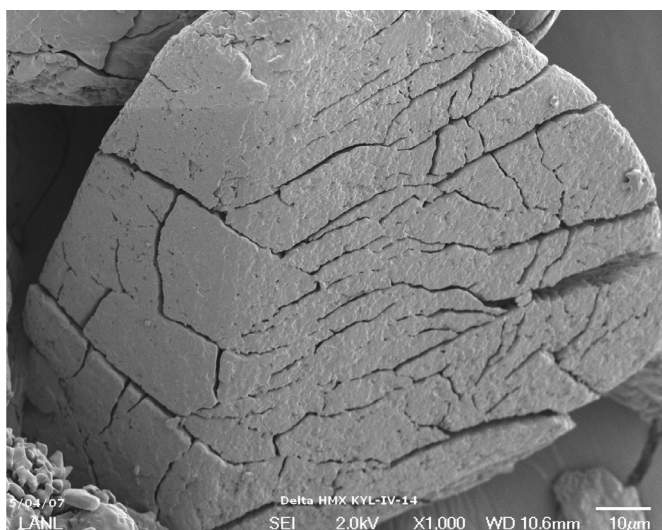


Fig. 1. Scanning electron micrograph of an HMX crystal, severely fractured from the β - δ transition caused by thermal exposure above 180 °C.

thermally damaged PBXs from granular high explosives (HE). We conclude by explaining how these observations may advance the physical descriptions that inform DDT mechanisms for implementation in state-of-the-art models.

2. Background

2.1. Thermal damage and cookoff

There is a longstanding need to understand the response of chemical high explosives exposed to abnormally elevated thermal environments during storage and transportation activities [1–3]. There have been many instances of HE accidents involving fire that resulted in violent explosions, sometimes with fatalities or significant financial costs. Thermal accident prevention has motivated research within the military, as well as at universities and regulatory agencies.

HMX is a crystalline, high-power explosive that is often formulated with polymer coatings to make PBX composites. Binders are added to convey desirable properties to the compositions. Principally, PBXs were developed to create materials that could be pressed to high bulk density—often approaching 98.5% theoretical maximum density (TMD)—and have enough structural integrity to maintain precise shape following subtractive machining operations. Furthermore, depending on the type of binder system used, the mechanical and thermal properties of the composition can be adjusted for increased safety. Whether all of the benefits derived from the binder system persist after thermal exposure to high temperatures, however, is not well known.

When an HMX-based PBX is heated, chemical and physical changes occur. Depending on the binder system used, a variety of endothermic processes such as melting of binder and vaporization of plasticizers may occur. Molten binder may become mobile and respond to stress from differential thermal expansion of other components and casing. HMX undergoes an endothermic β - δ polymorphic phase transition between 158–175 °C and can experience severe fracturing (Fig. 1) in individual crystallites as the lattice density decreases by 6.7% [4–6]. Eventually, solid-state decomposition reactions in both binder and HMX produce gas and porosity. If the PBX becomes permeable, a significant fraction of mass can escape through the porous network [7–10]. If strongly confined while heated, the expansion against the walls can lead to anisotropic expansion and non-randomly oriented frac-

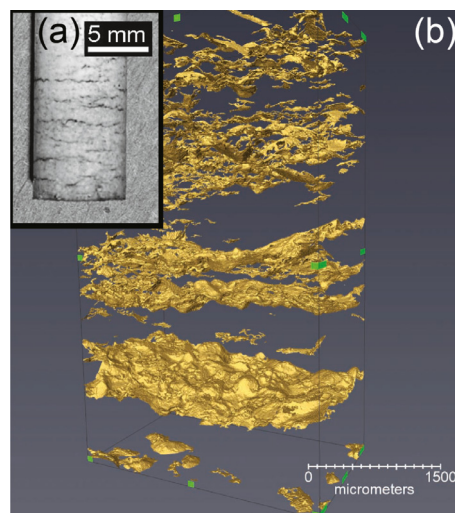


Fig. 2. Large field of view (a), and micro-CT (b) of aligned, normally oriented cracks in a cylinder of PBX 9501 thermally damaged inside a tube. The cylindrical axis is vertical in the images.

tures [11] (see Fig. 2). Depending on the boundary conditions that control thermal transfer to the environment and the reaction enthalpies in the energetic material being heated, self-sustaining, thermal runaway to ignition can occur. In the final stage of thermal runaway, exothermic, including autocatalytic, reactions begin to dominate, driving more heating and decomposition until an internal gas-filled cavity develops. The highly exothermic gas-phase recombination reactions associated with normal burning begin in this cavity, hence it has been described as an *ignition volume* [12,13]. This process by which a thermal exposure causes ignition is often referred to as *cookoff* or *auto-ignition*.

2.2. Deflagration modes

Deflagration in condensed-phase explosives is generally a surface regression process, independent of the ignition mechanism. It occurs when the bulk of the material is below the prompt decomposition temperature but the surface has been heated enough to drive off volatile intermediate products that recombine exothermically in the gas phase, producing a flame that feeds heat back to the surface, continuing the decomposition process.

For this to occur in cracks, pores or granular beds, there has to be space to accommodate the flame standoff distance, which decreases with pressure, otherwise the intermediate gaseous products are cooled by the adjacent surfaces before exothermic reactions can take place. If that condition is met, the local mass-burn rate is given by $\dot{m} = kAP^n$; where A is the burning surface area in a volume at the local pressure, P ; k is a constant of proportionality; and n is the burn rate exponent that is close to unity for HMX. Following ignition, flames spread across accessible HE surfaces. The flame propagation rate is controlled by the local pressure and the burn mode, which itself is described by the primary means of heat transfer (conduction, radiation or convection) from the flame reaction zone to the surface [14].

In contrast, if the bulk explosive is rapidly raised to a temperature high enough to produce rapid decomposition, which may happen radiatively or via strong compressive heating, then decomposition no longer proceeds via surface regression but rather bulk reaction, sometimes referred to as *thermal explosion*.

The rapid production of hot product gases, from either high-surface-area burning or bulk thermal explosion, creates intense pressure that works explosively on its surroundings and, when strongly confined, will further amplify the mass-burn rate [15–17].

The violence of these explosions, and the potential for increasing levels of damage or harm becomes particularly worrisome when the confinement conditions are sufficient to impel DDT. A review of the various burn modes, and terminology as used in this paper, is included below.

Conductive burning. Also known as *regressive* or *laminar burning*. Pressure-dependent, surface burning is sustained by heat conduction from the flame sheet to the condensed fuel, with propagation velocities on the order of 10^{-4} – 10^{-3} m s⁻¹. This mode will almost immediately become convective from pressure rise in a closed-end DDT tube when open porosity exists to accept infiltrating gases.

Convective burning. Hot gases infiltrate channels in porosity and transfer heat to the surfaces to initiate combustion; it is a reactive flow process. The velocity of a convective burn front is a function of the permeability of the porous medium, the temperature of the flowing gas and the pressure gradient driving flow. We assume gas-flow velocity within these spaces can propagate no faster than the local speed of sound in the gas. For the case of an ideal gas at 3200 K (the adiabatic flame temperature for burning HMX [18]) flowing in a straight unrestricted tube, we calculate an approximate flow velocity limited to $\approx 1000 \pm 100$ m s⁻¹, with uncertainty due to assumptions for adiabatic index ($\gamma = 1.34$) and mean molecular mass ($m = 0.03$ g mol⁻¹) of the combustion products of fully oxidized HMX. In real granular beds or low-porosity consolidated media, cracks and pores are tortuous, so that even with locally sonic-limited flow, the average rate of advancement of the convection front through the bulk will be much lower ².

Deconsolidative burning. Originally called *progressive deconsolidation* by Fifer and Cole [21], this is a mode of high-surface-area burning that follows an ignition precursor advancing in initially compacted HE beds. The primary characteristic is that particles (or multi-granular chunks as is the case with PBXs) detach and flow in the opposite direction from the burn advancement direction. The detached pieces are subsequently consumed in the flame plume. This is not a burn propagation mode, per se, rather it acts to increase the burning surface area and mass-burn rate. Deconsolidative burning is similar to convective, in some respects. Specifically, when the pressure in the flow behind the consolidated HE increases beyond a critical value, the flame structure becomes compressed to a sufficiently small length to be able to intrude into micro-scale voids, after which there is a sudden increase in accessible burn surface. The mechanism for flame intrusion was theorized by Belyaev et al. [22], with critical pressure thresholds measured by researchers for mechanically [23] and thermally [24] damaged PBX 9501. Closed bomb and strand burner experiments show rapid pressure rise, and chaotic burn progression, following the onset of deconsolidative burning in PBXs above the critical pressure [24–26]. Direct optical observation by Parker [11] showed deconsolidative burning in hot PBX 9501 damaged inside tube confinement. In that study, flames were seen intruding into periodic fractures oriented normal to the direction of the cylindrical axis (see Fig. 2), causing detachment of small chunks of PBX that traveled away from the back end of the initially compact PBX. Similar phenomenology has been observed in binderless pressings of HMX [21] and PETN [27].

Compressive burning. Compaction of a porous bed from compressive loading is inelastic and dissipative causing hot spots to form. As described by Frank–Kamenetskii theory [28] a critical hot spot will run away thermally and eventually ignite. The compressive burn front is a locus of spontaneous ignition (rather than ignition from reactive flow) where a population of critical hot spots mature to the ignition temperature in a kinetically deter-

mined induction time. As such, the compressive ignition locus approximately parallels the compaction wave front and can attain velocities on the order of the sound speed in the solid (e.g., ≤ 2.5 km s⁻¹ for hot PBX 9501 [29]). The induction time is a function of the dissipative work and chemical kinetics, and scales inversely with the force required to compact the bed, as well as the change in density across the compaction front. At higher densities, the force to cause compaction can be great [30–32] and the induction time short [33]; we call this *prompt compressive burning*.

Thermal explosion. This process for rapid ignition spreading is concisely defined by Hill [34] as follows: “A thermal explosion occurs when the temperature of a reacting energetic material spontaneously runs away. A thermal explosion becomes a physical explosion if the temperature runaway also causes the pressure to run away ... A thermal explosion produces a physical explosion only for pressure building systems, which are those for which the reaction products would seek to occupy a significantly larger volume than the reactants”, as is the case for DDT tube experiments filled with chemical explosives. Thermal explosions can be caused by: direct application of heat (i.e. cookoff); stimulated by compressive work (i.e. compressive burning); or produced eventually in a volume where high-surface-area burning is happening (i.e. convective or deconsolidative burning).

2.3. DDT in granular explosives

The mechanism for Type I DDT, as opposed to Type II, in granular materials has been refined over the course of many decades from various researchers at different laboratories ³. In fact, many of the important features and physical processes were identified well before the mechanism was classified as *Type I DDT* [37]. A thorough review of this mechanism and all its nuance is not included in this article, but the curious reader will find useful descriptions and bibliographies in the chapter by McAfee [38], and the papers by Baer & Nunziato [35] and Bdzil et al. [36].

Regardless of the means of ignition, either direct-thermal or mechanical by impact, Type I DDT requires the formation of a compacted section (density >90% TMD) that acts as a piston. In DDT tubes with impact ignition techniques, the formation of the compact is immediate. When the ignition technique is direct-thermal, the formation of a piston begins when and where the permeability of the solid medium is inadequate to accept intrusion of high-velocity gases produced from initial combustion. It has been shown that at this point convective burning ceases [22,39–41]. Historically, this piston has been called a *plug* [42,43], but new evidence [44] suggests this feature may not be plugging flow along the entire length of its interface with the inner surface of the confining tube (especially inside weaker confiners like polycarbonate tubes). We prefer to describe this feature as a *piston* and will do so throughout this paper.

The conditions that control the formation of a piston are specific permeability of the solid medium—which itself is a property of the size, amount, tortuosity and interconnectedness of the porosity—and the viscosity and velocity of the gas mixture flowing into the porous medium. Once the strength of the bed is exceeded, it will collapse through shear fractures nucleating at grain-to-grain contact points [36,45], or whole crystalline rearrangement, and a compaction wave will extend for the time period that pressure continues to be applied to the back surface. The pressure of the combustion products, generated from either convective or compressive burning, controls the acceleration of the piston, which in

² There are studies where supersonic flame velocity was reported, but these tests were done in clean-sided, engineered cracks [19,20] that do not resemble the morphology of the real damaged materials investigated in this study.

³ In many respects, there is consensus for this mechanism, but some disagreements also exist on some of the finer points in the process; this is particularly true for the multi-phase physics implemented in some commonly applied modeling theories [35,36].

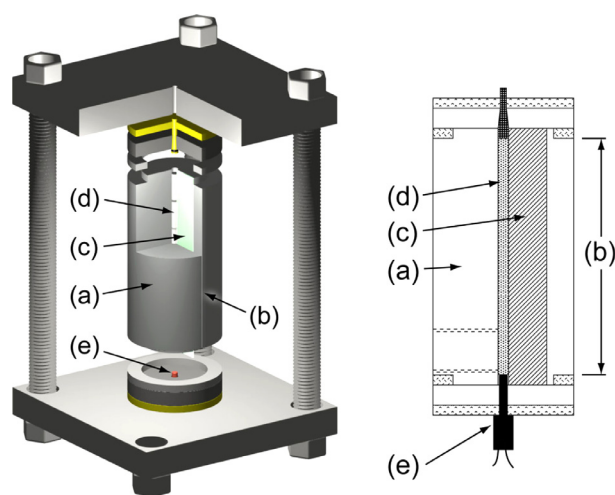


Fig. 3. Hot DDT tube experiment; (a) 38.1-mm-thick 1018 steel tube with a 0.5-mm slit (b) running lengthwise, (c) glass slide inside slit, (d) 6.35-mm-diameter $\times$ 150-mm-long explosive column, and (e) glow plug igniter.

turn controls the rate at which work is done in the compacting explosive. The impulsiveness of the burn propelling the piston should also greatly affect the rapidity of formation and strength of the eventual shock [46,47]. The rate of pressurization in the combustion zone is a function of pressure, accessible surface area (which is also related to the porosity) and the burn rate exponent of the burning material. It is worth emphasizing that acceleration of the HE piston, regardless of how it is accomplished, is essential for Type I DDT.

In unconsolidated, low-bulk-density granular beds, initially there is much porosity and surface area, high permeability and low bed strength. Convective burning can progress readily with sometimes significant penetration of gases before compaction starts. Once compaction begins, and convection ceases, there may be a sizeable volume containing residual burning particles in the following flow. Sometimes a rapid thermal explosion will occur in the convective burn infiltration region [48,49]; this further amplifies the force acting on the piston.

With consolidated PBXs, at higher densities and with their polymer binder systems, the initial conditions are considerably different. Intuitively, porosity and surface area are lower. The interconnectedness of the porosity is also less, creating low specific permeability in the medium [7,10]. Polymer binders are compliant and affect the bulk mechanical response, as well as interactions between HMX crystallites. Binders may also physically and chemically affect surface burning of coated explosive crystallites. Many of the controlling factors for Type I DDT are diminished or effectively eliminated and it becomes questionable whether DDT can even occur [50]. It is clear, that if DDT is possible in consolidated PBXs, details of the process progression are likely to be different. Identifying these differences and the mechanistic effects, if present, are the goals of the research in this paper.

3. Experimental methods

A DDT tube experiment was designed to survive heating to 250 °C (Fig. 3). The test consists of a heavy steel tube with a 0.5 mm-wide slit for optical access machined along the cylindrical axis. The tube was 152.4 mm long with an outer diameter of 82.6 mm and a 6.40 mm-diameter bore. Wire electrical discharge machining was used to cut the borehole and slit with ± 0.0005 -inch tolerance and parallelism to accommodate pressed HE cylinders in the bore and a precision-width, custom-length glass micro-

Table 1
Irreversible thermal expansion measurements for unconfined PBX 9501.

Sample	Initial		After 10 h. at 180 °C		
	L. (mm)	Dia. (mm)	L. (mm)	Dia. (mm)	Δ Vol. (%)
1	9.93	9.96	10.48	10.45	16.2
2	8.98	8.91	9.48	9.36	16.5
3	7.96	7.96	8.37	8.33	15.2
4	6.99	7.01	7.34	7.36	15.8
5	6.04	5.98	6.33	6.25	14.5
6	4.98	4.93	5.24	5.17	16.0
7	4.05	3.98	4.26	4.19	16.2
			Mean $\pm$ SD		15.8 $\pm$ 0.8

Table 2
PBX composition details.

PBX type	HMX fraction	Binder fraction and type	Plasticizer fraction and type
PBX 9501	95 wt-%	2.5 wt-% Estane 5703-F1	1.25 wt-% BDNPA, 1.25 wt-% BDNPF NA
LX-14-0	95.5 wt-%	4.5 wt-% Estane 5702-F1	NA
LX-10-1	94.5 wt-%	5.5 wt-% Viton A	NA
PBX 9012	90 wt-%	10 wt-% Viton A	NA

scope slide in the adjoined slit. The slide was not sealed within the slit, so it is assumed that gases produced along the length of the HE column during thermal decomposition could vent slowly to the environment. In contrast, during rapid combustion, product gas flow would be choked at the tight gap between slide and slit, effectively sealing the system. The slit was machined with width-reducing steps partially enclosing the outer edge of the slide to prevent the slide from being ejected during explosive burning. The ends of the tube were capped with polyether ether ketone (PEEK) insulating disks and steel plates pulled together with high-strength bolts.

HMX powder, both loose and uni-axially pressed into cylinders, and pressed cylinders of HMX-based PBXs were tested. Pressing allowed control of the initial bulk density for the explosive parts; initial density was measured with an error of ± 0.02 g cm⁻³. When loose HMX powder was used, bulk density was varied from either pour- to tap-density, or higher densities achieved by hand tamping with a wooden dowel. To make the density uniform, tamping was done incrementally each time a length equal to the tube diameter was filled with powder. Similarly, PBX compositions were pressed uniaxially to high and intermediate densities in a pressing die. At the lowest densities, PBX molding prills were poured into the tube. When pressed PBX cylinders were tested, it was necessary to shrink them in a liquid nitrogen bath before loading owing to the tightness-of-fit inside the DDT tube. Pressed parts were 6.35 mm high $\times$ 6.35 mm diameter. The parts were stacked end-to-end inside the tube and the length of the last part was trimmed to ensure the length of the bore was filled completely. Calculated ullage, due to the greater radius of the bore, was $\approx 1.6\%$. During heating of the DDT tubes, the HE undergoes expansion greater than the ullage inside the tube and, in doing so, closes the gap between its lateral surface and the inner walls of the confinement. Volumetric expansion measurements for unconfined PBX 9501 heated to 180 °C were made and are presented in Table 1. Composition details for the HE tested can be found in Table 2.

A custom-shaped foil heater was adhered to the tube along with 4 surface mounted thermocouples. The heater covered the entire length of the tube and almost all of its circumference, except where the thermocouples were attached and along the length of the slit to maintain optical access. The heater did not cover the

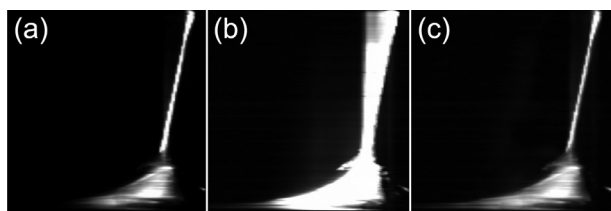


Fig. 4. An example of HDR image construction technique; (a) low-brightness image, (b) high-brightness image, and (c) merged HDR image. Images show DDT in β -HMX loaded at 19% initial porosity.

flat ends, where the assembly was clamped closed. Two additional thermocouples were inserted into the tube at both ends to measure the temperature of the explosive column. These thermocouples were potted in place with a high-temperature epoxy resin (JB Weld). The length of the tube was also overwrapped with insulation, with care being taken not to obstruct the slit, to improve the uniformity of the thermal boundary condition.

A PID thermal process controller, informed by a mid-length surface mounted thermocouple, was used to control heating rate and set-point temperature. Heating rates were kept within a range of $8\text{--}10\text{ }^{\circ}\text{C min}^{-1}$. End-to-end thermal gradients along the HE column were typically less than $3\text{ }^{\circ}\text{C}$.

For the cookoff experiments, the HE was heated to a high set-point and held steady until thermal runaway and auto-ignition occurred. Other tests were pre-heated, then end-ignited on the bottom side by a diesel engine glow plug threaded gas-tight into the DDT tube and abutting the HE. This means of ignition was chosen to best mimic the process of auto-ignition, which is purely thermal in nature. It differs from more impulsive means that have been used traditionally, such as impact by piston or thermal ignition by rapidly burning thermite or pyrotechnic mixtures.

Regardless of whether the HE was auto-ignited or end-ignited, the timing was somewhat unpredictable on the order of minutes or seconds, respectively, making it impossible to reliably capture image data with a conventional streak camera. Instead a method was developed to compose streak images from high-speed video frame sequences, the details of which are reported in an earlier article [51]. The advantage of this technique arises from the video camera having a significant period of pre-trigger recording allowing a post-ignition trigger signal to be used. Triggering was accomplished by transmitting light of reaction through a glass, polyimide-clad optic fiber, potted in a through-hole in the wall of the tube, to a photodiode being recorded on an oscilloscope.

A PhantomTM v710 (Vision Research, Inc.) 12-bit monochromatic camera was oriented to view through the slit for access to the lateral surface of the HE column. The dynamic range of light produced in a DDT event is great, starting with dim burning and progressing to very bright detonation. The 12-bit sensor in the camera was adequate to capture this range of luminosity, but because print media cannot reproduce the same contrast ratios, the streak images included in this paper are composites constructed in Adobe Photoshop CS6 software (Adobe Systems, Inc.) by merging a low-brightness streak image with a high-brightness image. This method for tone mapping pulls out details lost, respectively, from locally under- and over-exposed regions in each source image (Fig. 4). The resulting high dynamic range (HDR) images presented, reveal qualitative structures in the luminous reaction progress, but are not suitable for any type of quantitative analysis based on relative pixel luminance. Wave velocities can be measured from streak images, as well as the *DDT run length*, defined as the distance from the point of ignition to the initiation of a detonation wave.

After test completion, the damaged steel tubes were sectioned and examined. Typically, the transition to detonation was clearly

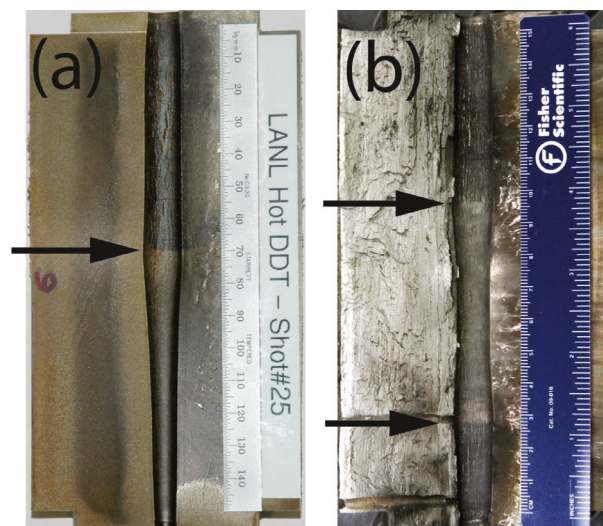


Fig. 5. Post-mortem images of sectioned DDT tubes from tests that were; (a) end-ignited by glow plug on bottom, (b) auto-ignited by cookoff. The arrows indicate the transition locations.

seen from the deformation profile along the inner surface of the tube (the tube would show increasing distension until a maximum at the transition region) and blackening from the transition point to the end of the tube (Fig. 5). This measure of DDT run length matches with the transition location in the streak images. When both image and tube run length measurements were made for a test, we report measurement uncertainty to be the lesser of the two methods. Typically, this is determined by the precision of the ruler ($\pm 0.5\text{ mm}$). The uncertainty of the streak image measurement is theoretically half of the pixel resolution ($\approx 0.3\text{ mm px}^{-1}$), but due to motion blur from a detonation wave moving some distance during the video interframe duration of $1\text{ }\mu\text{s}$, it can be difficult to accurately identify the detonation transition point. As a result, the positional uncertainty from image analysis can increase to be on the order of $\pm 1.0\text{ mm}$, or more depending on detonation velocity, thus this inferior method was rarely used.

4. Numerical methods: porosity model

Initial porosity, ϕ_i , is the void volume fraction and is calculated by the following equation:

$$\phi_i = \frac{(\rho_t - \rho_i)}{\rho_t}, \quad (1)$$

where ρ_t is TMD and ρ_i is the initial geometric density of the HE load. In hot HEs, porosity can come into existence in two ways: from the initial pressing, or evolving from the thermal damage. Porosity from pressing is an easily controllable experimental parameter, but the evolution of porosity from thermal damage is harder to determine accurately. There are a variety of reasons for this difficulty. First, porosity can evolve from the thermal decomposition and vaporization of both the binder system and the HMX. Secondly, porosity can be reduced when the HE is heated inside confinement when expansion of the HE is greater than the confiner. The expansion disparity can be significant with HMX-based HEs resulting from the β - δ phase change and volumetric expansion due to normal thermal expansion of the heated material.

Three major approximations are made for calculating the porosity evolution due to thermal damage. One, any gases or vapors produced by the thermally damaged binder or HMX will immediately vent to atmosphere. This assumes that the PBX and test fixture are highly permeable. In actuality, the test fixture is leaky owing to the

slit along its length, whereas the PBX has low permeability varying with its degree of porosity [7,8]. Two, the PBX will irreversibly swell to fill any free space in the DDT tube. This effect is shown for PBX 9501 in Table 1 and is assumed to be similar for the other HMX-based PBXs tested. Three, if there is not enough space in the tube to accommodate the phase change and thermal expansion of the PBX, then it is assumed that the porosity is zero.

The thermal decomposition of PBX 9501 is modeled by using a four-step kinetics model developed by Dickson et al. [52] for PBX 9501 to generate a temperature field over the full three dimensional volume of PBX. Then the thermal decomposition of individual components of the PBX 9501 (EstaneTM, nitroplasticizer and HMX) are calculated over this temperature field using a model for permeability development in thermally damaged PBX 9501 [53].

The final porosity, ϕ_f , is determined by calculating the mass of HMX and binder left after the thermal decomposition and then subtracting the volume change from both the HMX phase change and thermal expansion of the PBX composite. Next, the internal volume of the DDT tube assembly is calculated according to the thermal expansion of all the components. If the volume of the thermally damaged PBX is less than the volume of the expanded tube, the difference between the two is the porosity. If the volume of the thermally damaged PBX is greater than the volume of the tube, the porosity is assumed to be zero. The model assumes the components in the PBXs are able to rearrange, essentially flow, to access any free volume. We acknowledge that this is approximate. In reality, we expect it to be highly unlikely that the porosity could actually ever equal zero. It is more likely that the β - δ HMX phase change is inhibited, in this case, and instead, there will be some mixture of β - and δ -HMX, as well as some amount of porosity. We currently have no means to quantify the uncertainty described here, hence we report ϕ_f with error bars extending to the initial condition, assuming the actual value lies somewhere within the range. The data suggest that this approach produces a reasonable estimation for binning HE response based on porosity ranges.

This methodology for PBX 9501 can be extended to the other HMX-based PBXs with different binder systems. The decomposition kinetics for the binders used in LX-10 and PBX 9012 (VitonTMA) and LX-14 (EstaneTM5703) can be described by the flexible extended Prout-Tompkins nucleation growth model [54]. Since these will decompose differently in the presence of the reactive HMX, the coefficients for this model were refit to thermogravimetric analysis (TGA) data of the explosive using Markov Chain Monte Carlo (MCMC) to resample around the literature values of the kinetic parameters. This was done using the MCMC package in the Python programming package. After the new parameters were determined, the same methodology used for PBX 9501 was applied to determine the porosity of the other HMX-based PBXs of interest. The data and model results are shown in Fig. 6. This model was developed from TGA data taken at 180 °C and will likely become less accurate at temperatures higher than that, as was the case during thermal runaway in the cookoff tests reported herein. While the accuracy of the model at higher temperatures is unknown, we expect that this is not a significant error on the estimated bulk porosity because thermal runaway, by nature, is localized and the time-fraction of the thermal runaway stage was small relative to the total heated duration of the cookoff tests.

5. Results and discussion

5.1. Baseline response studies: DDT in HMX powders

To provide context with other DDT research done with steel tubes, we performed a series of baseline experiments with β -HMX powder, spanning a range of porosity from $\approx$ 1–50%, inside steel DDT tubes. We do not compare to literature data from the body

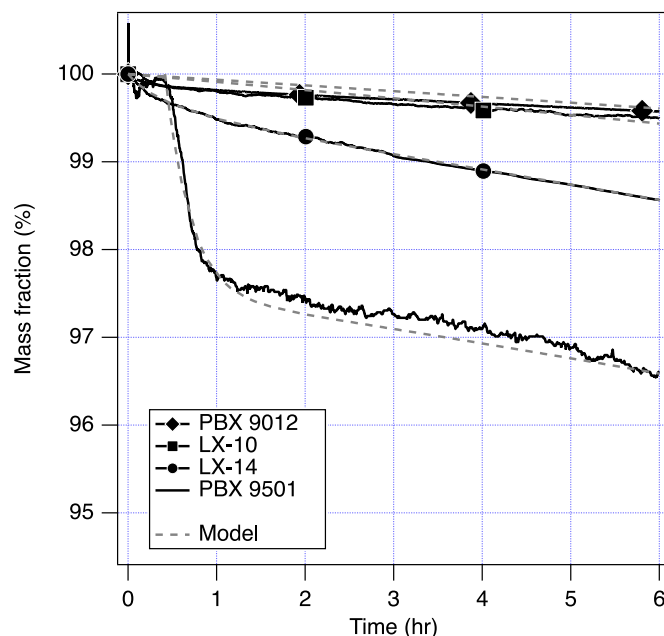


Fig. 6. Mass loss data and model from isothermal TGA at 180 °C for the four PBX compositions tested in this study.

of DDT work done with HMX powders inside polycarbonate tubes because the relative weakness of polycarbonate generally causes an increased run length [50,55], the effect of which can also be seen from the tests done in weaker brass tubes [56]. As is common in reporting DDT research, we use DDT run length⁴ as a measure of an energetic material's sensitivity to undergo detonation transition dependent on the amount of porosity (Figs. 7, 9, 10, 13 and 15). Run length, by itself, is only useful as an indicator of the integrated effects in DDT response. Consideration of more information-rich data (e.g. streak imaging) is required to gain insight into the interactions of different physical processes occurring within this length. Figure 7 shows our DDT run lengths compare well with other results [42,56–60], disregarding factors such as means of ignition, tube diameter and HMX particle size. The second-order polynomial least-squares fit to the data shows the concave-up curvature that is consistently seen by other researchers testing DDT with HMX and many other powdered forms of energetic materials [48,61–66]. Our data cover a larger range of porosity, including 2 tests performed near TMD with transition run lengths between 59–62 mm.

To account for and gain insight into the influence of the HMX β - δ phase transition, we performed a series of tests, spanning a range of porosity, with room temperature β - and δ -HMX. The δ -HMX was prepared by heating loaded DDT tubes to $\approx$ 185 °C for 1 h, then cooling slowly to room temperature over a period of $\approx$ 14 h. The respective polymorphs were confirmed by Raman spectroscopy with results indicating the thermally treated HMX was predominantly δ -phase (Fig. 8). Consistent with the spectrographs of Tappan et al. [67], we observed the loss of the doublet in the 420 cm⁻¹ region and a downfield shift of the 3-peak group in the 900 cm⁻¹ region of the β -HMX spectrum, combined with the appearance of a 450 cm⁻¹ peak in the δ -HMX spectrum. These

⁴ This is done for practical purposes; specifically, for risk assessments made by regulators and design engineers, many scenarios-of-concern for accidental detonation of explosive-containing devices with known dimensions can be eliminated based on whether the device is large enough for the DDT process to reach completion.

⁵ It is likely that some fraction of residual β -HMX exists, but the ratio is unknown.

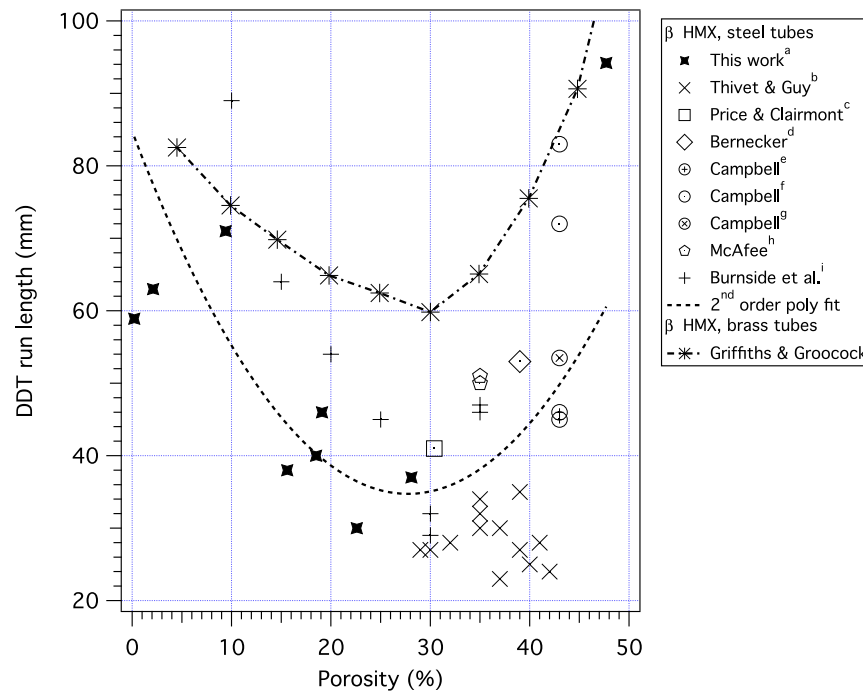


Fig. 7. Data compilation of DDT run length as a function of ϕ_i for granular β -HMX in metal tubes at room temperature. Note: the Griffiths and Grocock data are not included in the least squares fit. HMX particle size and tube inner diameters (ID), respectively, are: (a) 3:1 bimodal, mean = 170.44 μm , ID = 6.35 mm; (b) 0–100 μm , ID = 1.2 mm [57]; (c) mean = 170 μm , ID = 16.3 mm [58]; (d) very coarse, ID = 25.4 mm [50]; (e) mean = 170 μm , ID = 6.35 mm [59]; (f) <44 μm , ID = 6.35 mm [59]; (g) calculated length by 3:1 weighting of run lengths from e and f [59]; (h) mean = 170 μm , ID = 12.7 mm [42]; (i) mean = 170 μm , ID = 6.35 mm [60]; (j) 124–251 μm , ID = 12.8 mm [56].

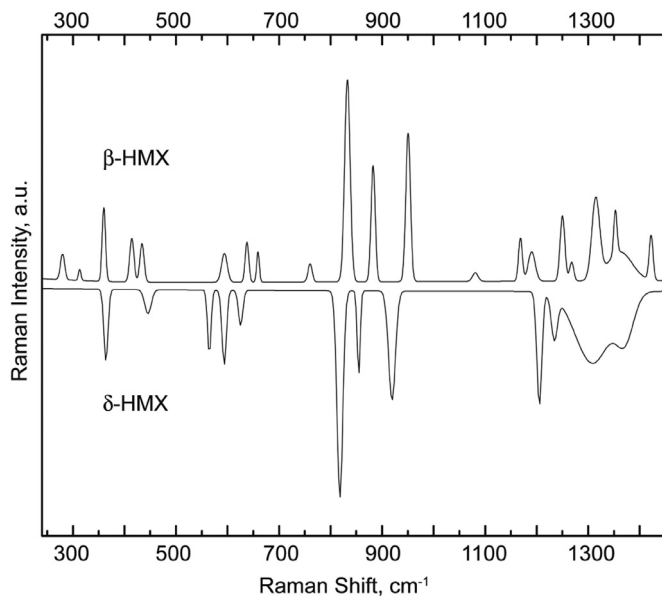


Fig. 8. Raman spectrographs of HMX used in this study.

changes are characteristic of thermally induced δ -HMX polymorphs emerging in initially β -HMX crystal population. Additionally, we explored the combined effects on DDT run length of the δ -HMX at elevated temperature, by testing at 185 °C following a 1-hr soak duration; Fig. 9 shows these data. Nearly all of the HMX-loaded tubes transitioned, with the sole exception being β -HMX pressed to $\approx 95\%$ TMD; this sample only deflagrated. It appears that the δ -HMX is possibly more sensitive than β -HMX in the sense that the run lengths for DDT were shorter in all trials except two. The trial of hot δ -HMX at $\phi_f = 17.5\%$ was unusual, in respect to the other HMX powder trials, because it transitioned to detonation in

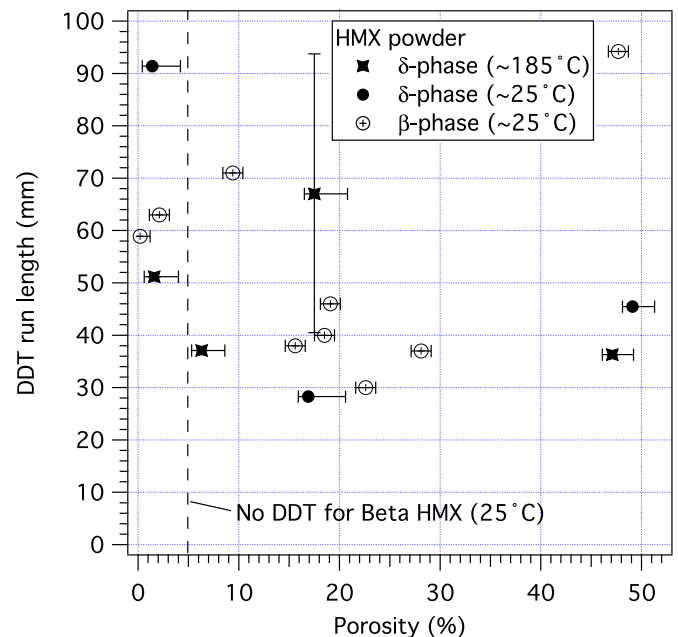


Fig. 9. Data showing the effect on run length across a range of porosity for β - and δ -phase polymorphs of granular HMX at elevated and room temperature. Unless shown, uncertainty on run length is ± 0.5 mm.

both directions (see streak in Fig. 12). The error bars for this data point are large as a result of there being two ways to measure the run distance: the first is to measure from the ignition end of the tube (as is generally done for end-ignited tests), the second is to halve the distance between the two transition locations (as is done for auto-ignited tests). Even though this test was end-ignited, the thermal explosion resembles what is seen in the auto-ignited trials. The uncertainty makes either measurement method reasonable and the error bars capture this.

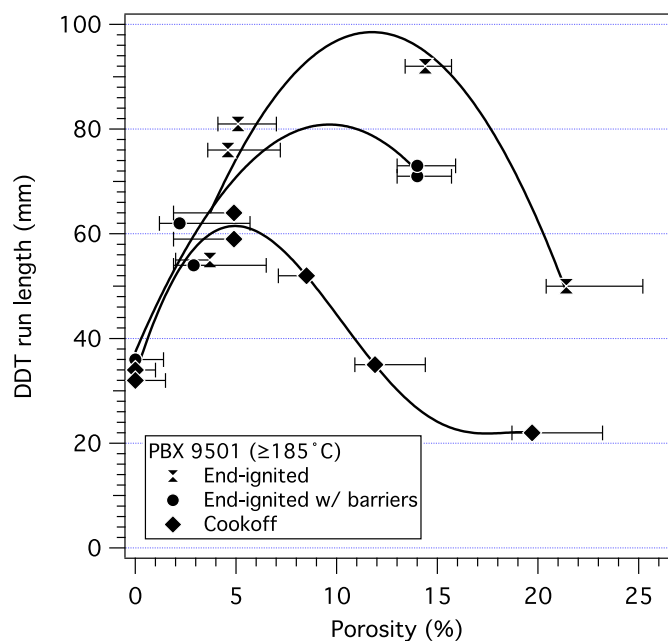


Fig. 10. Data for all hot PBX 9501 trials showing concave-down trend and run length convergence at the low range of ϕ_f . Data fit by least-squares polynomial regression. Uncertainty on run length is ± 0.5 mm.

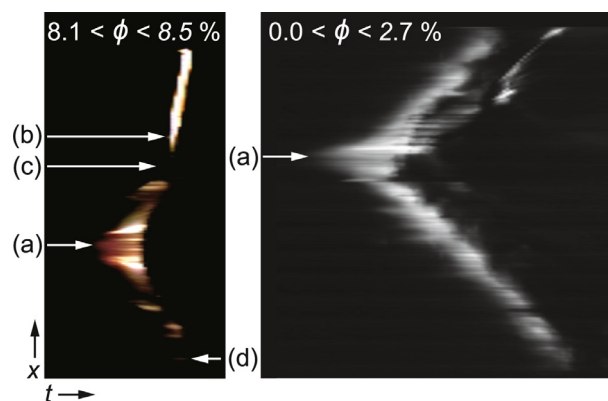


Fig. 11. Example streaks of auto-ignited PBX 9501 (left) and PBX 9012 (right) trials showing symmetric reaction growth. Reaction spreads at ≈ 1.5 – 2.5 km s^{-1} from the ignition location (a) to the transition to detonation (b). The compacted HE piston, (c), is identifiable by diminished luminosity in this region. In this trial, reaction reached the bottom end of the tube, (d), before transition could be completed in this direction.

5.2. Baseline response studies: Cool PBX 9501

PBX 9501 was tested in both pristine and damaged states. Three trials with pristine, room temperature PBX 9501 molding prills, pressed to 98% ($\phi_i = 2\%$), 93% ($\phi_i = 7\%$) and 86% ($\phi_i = 14\%$) TMD, were end-ignited and did not transition to detonation. However, when PBX 9501, pressed to $\phi_i = 2\%$, was thermally cycled

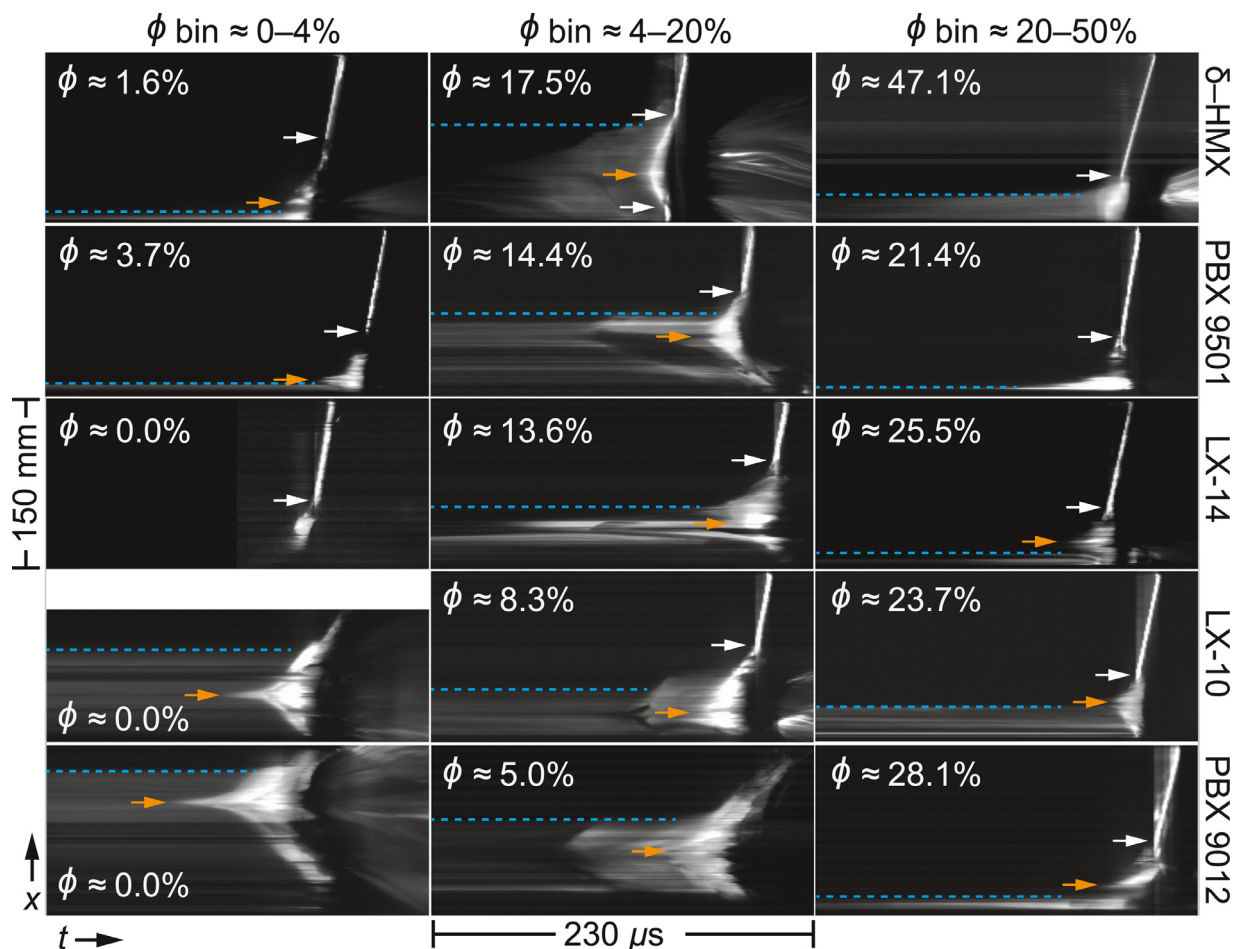


Fig. 12. A sample of streak images (end-ignited on bottom) for the four PBXs and δ -HMX powder tested hot, binned by ϕ_f . White arrows indicate transition location, orange arrows indicate thermal explosion (when it can be identified), and blue dashed line indicates flame infiltration depth.

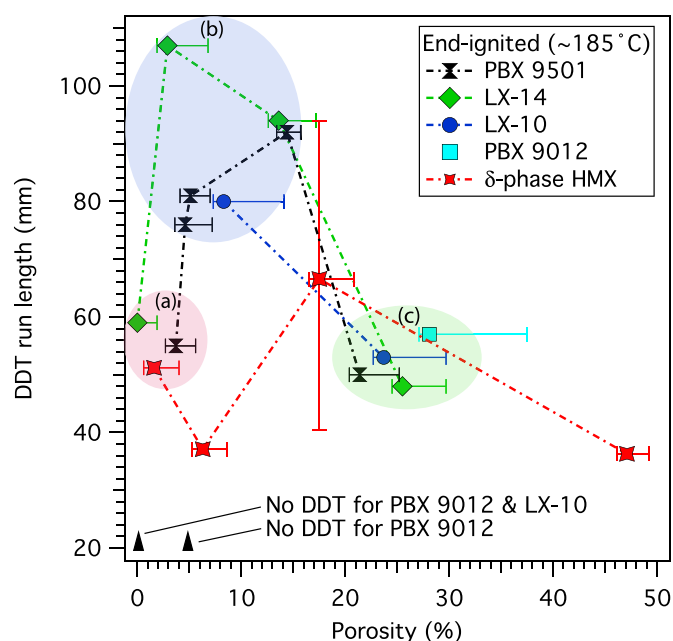


Fig. 13. Data for all end-ignited, hot PBX trials. PBX 9501 and LX-14 show a concave-down trend. Run lengths converge at the lowest range of porosity. Also shown for comparison are data for hot δ -HMX powder. The porosity bins referenced within this paper are grouped as follows: (a) low- ϕ_f bin, (b) intermediate- ϕ_f bin, and (c) high- ϕ_f bin. Unless otherwise shown, uncertainty on run length is ± 0.5 mm. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

above 180 °C and cooled back to room temperature ($\phi_f = 5\%$), DDT occurred in a run length of 90 mm, demonstrating that the irreversible morphological changes from thermal exposure above 180 °C are necessary to permit DDT. Figure 2a shows post-mortem images of a cylinder of PBX 9501 thermally damaged and cooled inside a tube. Cracks aligned normal to the cylindrical axis are apparent on the lateral surface and are shown to extend through the sample in the x-ray computed tomography (Fig. 2b). Earlier micro-structural characterization efforts examining PBX 9501 exposed to the same thermal cycle inside a laterally confining tube also showed these periodic cracks as well as a reduction in HMX particle size [8,11]. The fact that undamaged samples with similar or greater amounts of porosity did not transition, but the damaged sample did, demonstrates: (1) the morphological qualities of porosity created by thermal damage, and (2) the presence of more sensitive δ -phase HMX, are more important than the quantity of porosity, in affecting the progression of explosive violence. This is not the case for room temperature powders.

5.3. DDT behavior in Hot PBX 9501

Three variations of DDT tests were performed at elevated temperature with PBX 9501. The tests in the first series were heated to approximately 185 °C for a duration of 1–3 h, then end-ignited. For the second series, 0.5-mm-thick, 6.35-mm-diameter PTFE flow barriers were emplaced between HE cylinders at three locations near the ignition end of the HE column. As in the previous series, these tests were heated to the same temperatures for the same soak duration. The third series of tests were cookoff trials. The samples were heated to approximately 185 °C and soaked for 1 h to accumulate damage, then the heat was increased at a rate of 8–10 °C min⁻¹ until auto-ignition occurred. In each series, the PBX 9501 initial density was varied to span a range. The DDT run length data from these tests are combined in Fig. 10.

Regardless of ignition mode, transition to detonation in hot PBX 9501 happened with each trial and, when duplicate trials were run, the DDT run lengths were reasonably reproducible. There are two insightful trends revealed in the least squares polynomial fits to the data. Most notable is the concave-down curvature. This is a departure from trends published by other researchers [48,56,60–66] and also displayed in Fig. 7, where the curve is concave-up with minimum DDT run lengths at intermediate ϕ_f . We emphasize that the concave-up trend frequently reported in the literature, was derived from powdered HE experiments, not thermally damaged PBXs, and was typically assumed through extrapolation to apply the lowest porosities. Unfortunately, experimental data from other researchers at low ϕ and with PBXs are sparse. The second important observation from our data is that at the lowest porosities tested, the most relevant to real cookoff accident scenarios with consolidated PBX charges, the run lengths converged for the three test series variants.

The convergence at $\phi_f < 4\%$, both with and without impermeable flow barriers in the HE column, suggests the PBX 9501 in these trials was equally impermeable and, therefore, the DDT mechanism did not depend on significant convective burning past the location of the first barrier (25.4 mm). Consistently, at higher porosity, i.e. $\phi_f \approx 15\%$, where it is plausible to assume greater permeability, the inclusion of flow barriers exhibits the effect of shortening the run length, demonstrating the run length may be strongly dependent on how promptly convective infiltration shuts down and compaction begins. This observation supports the supposition that at intermediate levels of porosity a transition between convective-burn-driven vs. compressive-burn-driven mechanisms operates, and the extended run lengths may be the consequence of disorganized, possibly competing, burn propagation modes at the start of the DDT process before the establishment of a thermal explosion.

Finally, for the PBX 9501 tests that were heated until cookoff, the ignition location was generally near mid-length of the tube, with luminous waves spreading symmetrically towards the ends (Fig. 11). Centralization of the ignition volume is typical for slowly heated charges [12,13] for reasons discussed in the Background section. Following auto-ignition, the DDT run lengths at the lowest ϕ_f were similar to the end-ignited tests. The cookoff trials also showed a concave-down trend for run length with respect to porosity, in the range of ≈ 0 –20%, resulting from run lengths being longer for intermediate ϕ_f samples. At intermediate and higher ϕ_f , the data suggest the thermal explosion from auto-ignition is more impulsive, and the Type I mechanism more spatially condensed, than the process that develops from ignition by glow plug at the end of the tube. In these trials, there is no indication of a slow (or fast) flame infiltration phase preceding the establishment of thermal explosion and bed compaction. The shorter run lengths in auto-ignited samples, compared to the end-ignited trials, could also be influenced by enhanced thermal sensitization, as the thermal runaway locally elevates the HE temperature above the imposed boundary temperature.

5.4. DDT in other HMX-based PBXs

The effects of the binder system used in different thermally damaged HMX-based PBXs were also examined. There were many similarities to the trials with PBX 9501, but also some important differences⁶. LX-14 and LX-10 are two PBX compositions with approximately 95 wt-% HMX fill fraction, like PBX 9501, but with different binders. Comparisons between these formulations elucidate the influence of evolved porosity in the binder field, as well

⁶ Fewer tests were performed with the other PBXs due to limited HE stock.

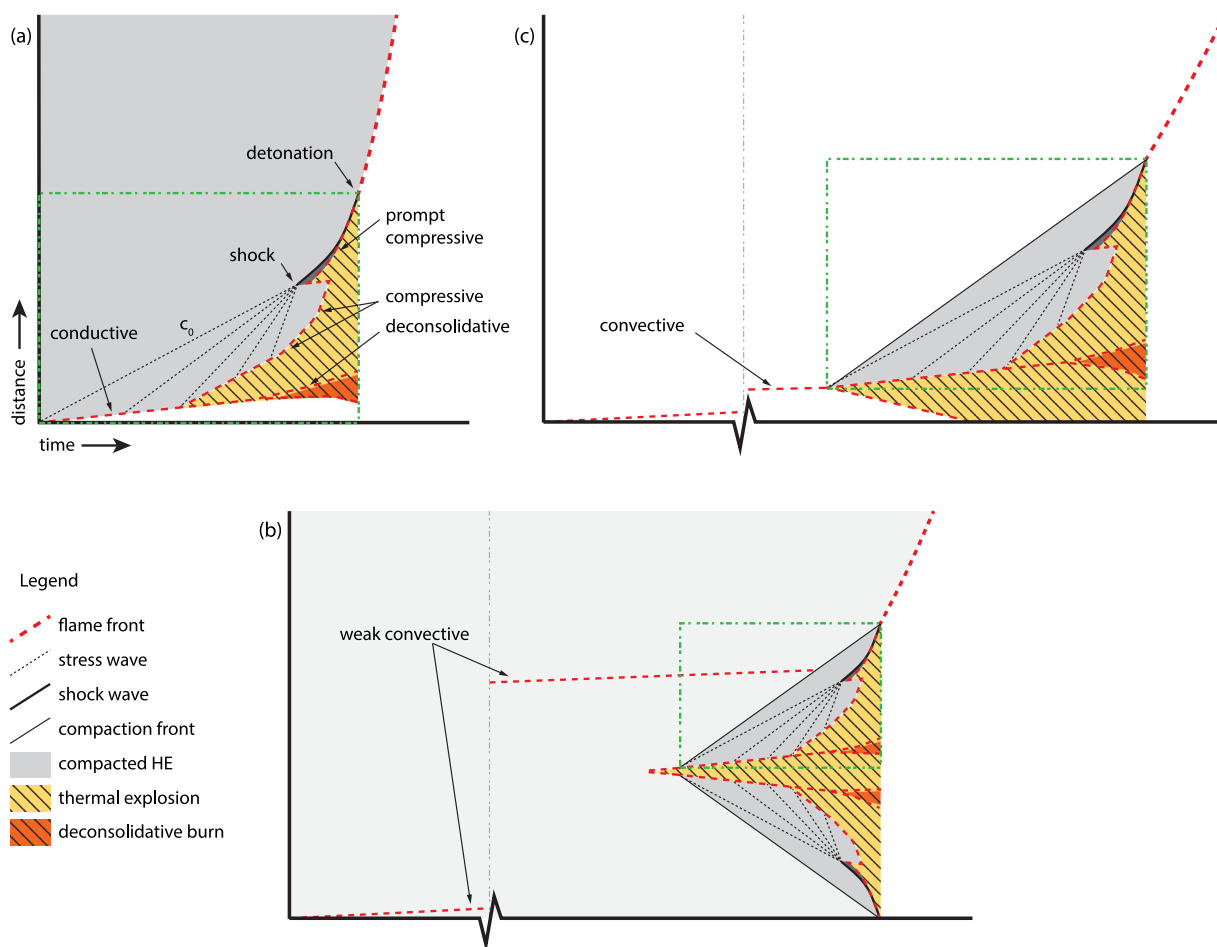


Fig. 14. Schematic representations of DDT mechanisms in hot, thermally damaged PBXs at low ϕ_f , (a), intermediate ϕ_f , (b), and high ϕ_f , (c). The green dashed box shows conserved features from Type I DDT. In each, a thermal explosion event arising by different means, drives acceleration and compaction/compression of the HE column.

as how thermal resilience of the binder can affect DDT propensity. Figure 6 shows isothermal TGA at 180 °C of the four PBX compositions used in this study. As is displayed, the binder system has a significant influence on mass loss and porosity evolution. LX-14 uses Estane binder, similar to PBX 9501, but lacks the BDNPA-F plasticizer which vaporizes at approximately 120–150 °C [53]. As a result, less porosity forms in the binder field during thermal exposure above 180 °C. LX-10 is bonded with the thermally stable fluoropolymer, Viton A, so the only porosity that forms during exposure at 180 °C comes from decomposition of the HMX component resulting in even less porosity formation. Lastly, we explored the effect of binder mass fraction by including PBX 9012 in the test series. PBX 9012 has twice as much Viton A, by weight, as LX-10. As with LX-10, little porosity forms in PBX 9012 from binder decomposition or vaporization.

We hypothesized that the presence of compliant binder systems would act to increase run length by reducing the efficiency of a compressive-initiated burn propagation mode. The binder component would dissipate a fraction of the available kinetic energy from the compressive loading, leaving less to form hot spots by intergranular interactions, specifically shear deformation, of HMX crystals. We hypothesized that binder would also shorten the distance of convective burn infiltration by fouling gas-flow paths between crystals and reducing permeability. Binder coatings on HMX crystal surfaces could also reduce the amount [25] of surface burning. Altogether, the deleterious influence of binders on convective burn propagation rate and pressure production would act to extend run

length. It was also hypothesized that these effects would be more evident in PBXs with a higher binder load and more thermally resilient binder formulations, by further extending run lengths, or eliminating the occurrence of DDT altogether.

Figure 12 contains the streak image data for each PBX, along with the data for neat δ -HMX at ≈ 185 °C to accentuate the effects of binder. Figure 13 displays the associated DDT run lengths. For this discussion, patterns in response are grouped into three porosity bins: low- $\phi_f \approx 0$ –4%; intermediate- $\phi_f \approx 4$ –20%; and high- $\phi_f \approx 20$ –50%.

5.4.1. Streak image data set

Interpretation of streak images can be challenging because, at first glance, there is considerable variation in appearance between trials. However, with experience, there are also certain features that can be identified in a majority of the results. In Fig. 12, we identify two important features, the flame infiltration distance (denoted by the blue dashed lines) and the thermal explosion (denoted by the orange arrows). The flame infiltration distance is a low-luminosity region that advances from the ignition end at relatively low rates (on the order of 10^{-2} – 10^1 m s $^{-1}$). The infiltrating burn propagation is the result of forced convection of hot gases traveling through cracks and causing surfaces to ignite along the way. The rate is sufficiently slow that it often appears to be flat relative to the faster, higher luminosity propagation modes that become evident ≈ 50 – 100 μ s before transition to detonation. The thermal explosion is characterized by an increase in luminosity

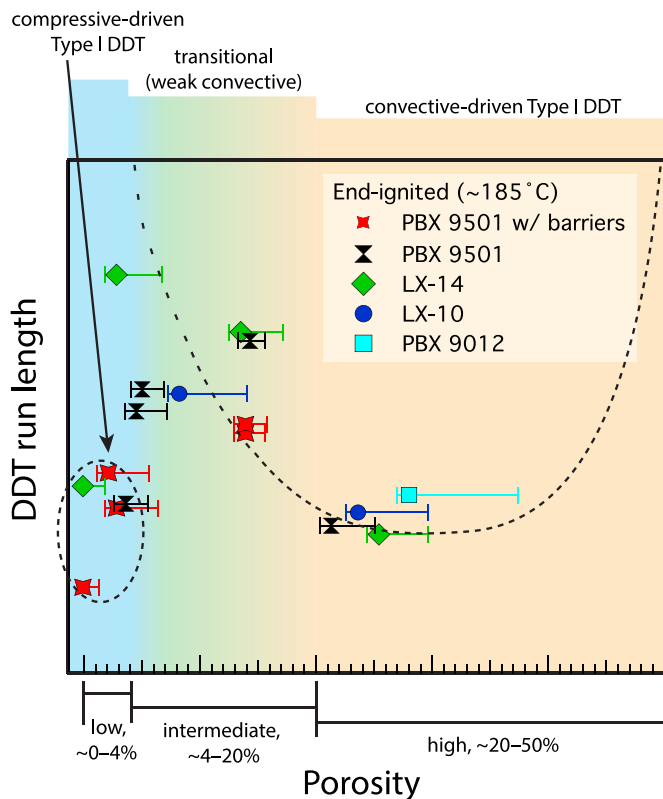


Fig. 15. Classification of Type I DDT in thermally damaged PBXs. Dashed lines are notional. The differentiation relies on the burn mode responsible for establishing a thermal explosion.

that propagates at $\approx 1.5\text{--}2.5 \text{ km s}^{-1}$ and frequently connects to the detonation transition location. This rate is too high for convective flow, but in the range expected for the speed of sound in hot PBXs [29], spanning a range of porosity, therefore we consider this to be a compressive burn mode. Usually the thermal explosion spreads in both directions from its origination point, creating a high luminosity feature that resembles the shape of a chevron. There is frequently a dimmer or dark region between the thermal explosion and the break-out of detonation; this is interpreted to be the compacted HE piston. Quintessential examples of thermal explosion are shown in Fig. 11, streaks of PBX 9501 and PBX 9012 during thermal auto-ignition. The PBX 9501 transitions to detonation, while the PBX 9012 does not progress past compressive burning.

Examination of the streaks for which DDT is observed shows other qualitative trends. In the low- ϕ_f and high- ϕ_f bins, the flame infiltration distance is less than or nearly equal to the distance at which the thermal explosion originates (when identifiable). The flame infiltration distance is also not as far into the tube as is seen with intermediate- ϕ_f or non-DDT tests. This suggests that compaction and the associated restriction of convective flow occurred promptly and the thermal explosion initiated at, or near, the back of the compacted HE piston. Considering the vast differences in initial conditions between high- ϕ_f and low- ϕ_f bins, specifically the amount of porosity and burn-accessible surface area, this is a surprising observation. In the intermediate- ϕ_f bin and for all non-DDT results, the thermal explosion occurred behind the flame infiltration front.

5.4.2. Low- ϕ_f bin

In the low- ϕ_f bin, the run lengths are similar for PBX 9501, LX-14, and neat δ -HMX (Fig. 13a); and DDT does not occur for LX-10 and PBX 9012. From this we conclude that Estane-based binder

systems have little effect on the DDT mechanism operating in this bin, whereas the Viton-based binder system does. The addition of thermally stable binders, like Viton A, clearly convey some benefit for mitigating thermal accidents where DDT is an unacceptable outcome. However, examination of the streak images for these tests when DDT did not occur (Fig. 12), reveals features that suggest the mechanism for Type I DDT was beginning to organize in a manner reminiscent of that seen in the intermediate- ϕ_f bin, but here the tubes were not long enough to allow the mechanism to progress to completion. Specifically, the chevron-shaped region we interpret to be a thermal explosion and common pre-cursor to DDT, is seen originating farther up the tubes in the tests with Viton-bonded PBXs. Ultimately, the effect of using a thermally resilient binder might be to protract, though not necessarily eliminate, the DDT mechanism in larger charges than tested here. Regardless, there were not enough trials performed to assess this benefit in a statistically meaningful way, or to confidently describe the physics for why this seeming benefit emerges. Further experimentation is needed.

5.4.3. High- ϕ_f bin

The PBX samples in the high- ϕ_f bin (Fig. 13c) have similar run lengths to PBXs in the low- ϕ_f bin, but are $\approx 25\text{--}50\%$ longer than unconsolidated δ -HMX powder at high- ϕ_f , showing the binder has some effect when porosity is high. As hypothesized, the binder causes the run length to increase, other than this, the rest of the DDT process appears to be qualitatively similar to what is typically seen in HMX powders.

5.4.4. Intermediate- ϕ_f bin

In the intermediate- ϕ_f bin (Fig. 13b), the run lengths are the greatest, $\approx 125\text{--}200\%$ longer than unconsolidated δ -HMX powder. As with the low- ϕ_f bin, DDT did not occur in PBX 9012.

5.4.5. Mechanisms

In this section, we draw from previous observations in the literature and inductive reasoning to propose explanations for the trends we have presented. In general, regardless of the amount of porosity, the final stages of the DDT process resemble what is seen in other examples of Type I DDT. That is, they appear to involve rapid acceleration and compression of a compacted HE piston. The differences, where they exist in our data set with PBXs, are in the early deflagration stages and how a propulsive thermal explosion is generated to accelerate the HE piston.

For the high- ϕ_f cases, gases can infiltrate the bed efficiently and high-surface-area convective burn mode can form an impulsive thermal and physical explosion, causing the start of compaction, in short (a few to $\approx 20 \text{ mm}$) infiltration distances. This porosity bin is the domain of classic *convective-driven* Type I DDT (Fig. 14c). Hot PBXs in this bin follow well described phenomenology and the observations made on DDT in room temperature HE powders appear to apply here. If the presence of binder is significant, it may be in that it coats crystals and retards the surface regressive burn rates to some degree.

With low- ϕ_f , the correlation of low permeability (manifested as restricted flame infiltration) with low levels of porosity, is both intuitive and experimentally supported [7,10]. The initial condition of the bed is compacted, so convective burning, as it exists in high- ϕ_f beds, should not occur. Another high-mass-burn rate mode is needed to produce the thermal explosion required to propel the HE piston. We propose compressive and deconsolidative burn modes to satisfy this criterion, hence we consider this bin to be the domain of *compressive-driven* Type I DDT (Fig. 14a). A compressive burn mode can propagate at the same rate as the compression wave in the hot full-density PBX ($\approx 2.5 \text{ km s}^{-1}$). Indeed, the propagation rates preceding detonation are on that order. Once ignition

sites form, the local pressure generated from burning at each site, may also force the bed to deconsolidate and expose additional surface area to be incorporated in the burn [21,24–26]. Compressive and deconsolidative burn modes are not mutually exclusive. While not necessary, both modes operating in concert could feasibly produce a rapid thermal and physical explosion in an initially compacted bed.

The responses of the low- ϕ_f Viton-bonded PBXs, however, challenge this interpretation framework and high-confidence explanation eludes us. Those tests show evidence of slow flame infiltration followed by thermal explosion within the infiltration zone. The initial rate of flame spread is too slow (10^{-2} – 10^1 m s $^{-1}$) to be explained by compressive burning. The deconsolidative burn mode is also not likely, at least early on, because this burn mode can only follow an ignition front that was advanced into the HE by some other mechanism. In the absence of evidence for a compressive precursor burn locus to initiate deconsolidation, we are left with convection to explain the observation of slow flame infiltration. Even though the estimated porosity is zero⁷, the streaks imply there must actually be a sufficient number of tortuous flow-paths for weak flame-producing reactions to travel slowly through the HE, or along gaps between the HE lateral surface and the inner surface of the tube. The origination point of thermal runaway to explosion, in these cases, likely results from stochastic localization of pressure and temperature anywhere behind the infiltration front due to the morphological heterogeneity in the thermally damaged composite.

The intermediate- ϕ_f tests (all PBXs) show similar phenomenology to the low- ϕ_f Viton-bonded PBXs, implying there is adequate exploitable porosity here to permit slow, non-vigorous convective burn propagation for durations of several milliseconds before onset of thermal explosion. This weak convective precursor appears to seed the HE bed with enough extra heat to cause thermal runaway somewhere behind the flame infiltration front. In this bin, for all test HEs except PBX 9012, the thermal explosion produced a physical explosion that drove compaction and compressive burning that overtook the flame infiltration front before transitioning to detonation. It follows that the reason for DDT run lengths being longer in this bin is due to the extended time and distance required to produce the thermal explosion. As is the case with the other bins, the late-time features following thermal explosion tend to resemble what is seen in streaks of Type I DDT. While this bin is somewhat transitional between the convective- and compressive-driven mechanisms, the nature of how the thermal explosion is generated aligns more closely with the high ϕ_f bin, hence we consider this to be a weaker flavor of the convective-driven type (Fig. 14b). Lastly, it is apparent from these data that an intermediate- ϕ_i bed of loose powder at room temperature is a decent surrogate for hot, thermally damaged PBXs at similar bulk density.

6. Conclusions

We designed a strong, high-temperature DDT tube experiment and used it to evaluate the behavior of hot, thermally damaged HMX and HMX-based PBXs with a primary goal of identifying differences in the mechanistic evolution of DDT between these and lower-bulk-density HMX powder beds. Further experimental developments permitted observation of unpredictably timed cookoff events and subsequent reaction propagation; these data are directly applicable to assessing evolution of reaction violence for PBXs in thermal-ignition (cookoff) accidents.

We demonstrated that the δ -phase polymorph of HMX is probably more sensitive (i.e. transitions to detonation in shorter dis-

tances) and that the sensitization is similar regardless of whether the experiment was performed at room temperature or at 185 °C. The observation of DDT in hot PBXs at high density ($\geq 96\%$ TMD), is also novel and may be due to high temperature sensitization of the HMX. In most of the tests, a thermal explosion was observed, generating a fast (≈ 2.5 km s $^{-1}$) bright wave connecting to the detonation transition location. The effect of a thermal explosion on the DDT process is clearly shown in the auto-ignited cookoff trials, which by definition are thermal explosions, where the transition took on the characteristics of Type I DDT, and did so in the shortest run lengths measured.

Perhaps the most important practical findings of this work concern the influence of binder systems. First, the addition of the thermally resilient fluoropolymer, Viton A, offers safety benefits in the sense that it either reduced the propensity for DDT, or increased the run distance required for the transition to occur in LX-10 and PBX 9012 beyond the length of our experimental apparatus. A higher Viton binder mass fraction seemed to impart extra benefit. Second, the observation of a concave-down trend in the variation in DDT run distance with respect to porosity indicates the wide-ranging DDT behavior of hot, thermally damaged PBXs was not captured in earlier work done with low-density, room temperature powder beds. In fact, a single concave-down trend line implies mechanistic continuity, where in actuality it seems likely two separate-but-related mechanisms are at play, differentiated by how a thermal, and associated physical, explosion is established. We believe it is more accurate to conclude, given the seemingly conserved qualities in the streaks following the onset of compaction, that there are two nuanced versions of the same Type I DDT mechanism operating: i.e. *convective-driven* Type I at high- ϕ_f , with a protracted version of the same at intermediate- ϕ_f ; and *compressive-driven* Type I at low- ϕ_f . If the intermediate- ϕ_f and high- ϕ_f bins are considered to be essentially of the convective-driven flavor, they can be combined and a single concave-up curve fits these data, consistent with the observations from DDT tests in the literature, done with room temperature granular explosives. It follows that the group of run lengths in the low- ϕ_f bin is the result of the compressive-driven flavor of Type I DDT and should be classified separately. Altogether, Fig. 15 illustrates a better way to classify our interpretation of these trends.

The observation of reproducible Type I DDT phenomenology for PBX 9501 and LX-14 at both low- ($<4\%$) and high-porosity (≈ 20 – 50%) suggests that application of continuum DDT models may be useful. However, before these can be successful, a considerable amount of research needs to be performed to measure high-temperature constitutive relations for these PBXs at a variety of relevant strain rates, as well as describing equations of state for porous media and reactive flows.

The challenges for modeling the DDT response for hot-thermally damaged PBXs at intermediate-porosity (≈ 4 – 20%) are greater. In this regime, a meso- and micro-scale characterization of thermal damage morphology, and the effects of burn propagation, will likely be required, at least when the material is ignited by a slow thermal source, as was done in this work. The seemingly stochastic nature of flame infiltration through fractures, prior to thermal explosion and the associated establishment of a strong compaction wave, may be eliminated by igniting the PBX column by more propulsive means, such as a metal piston impact. Piston-ignited DDT experiments with hot, thermally damaged PBXs are currently being designed to test this hypothesis.

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⁷ As discussed in the Numerical Methods section, the estimation of ϕ_f is likely incorrect with bias towards underestimation.

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